

# ECON 164: Theory of Economic Growth

## Week 8: Technology Adoption and Trade

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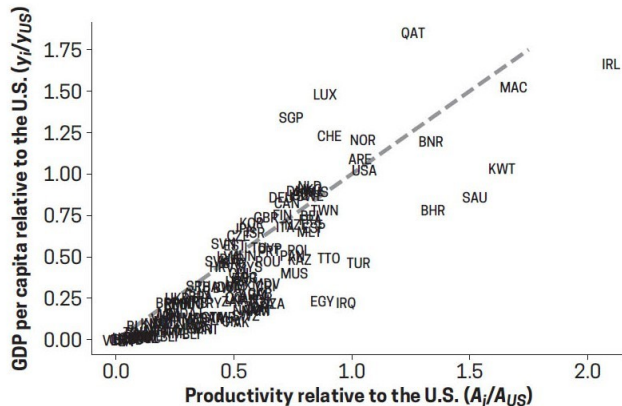
Levi Crews

Spring 2026

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  - share aggregate dynamics in which  **$A$**  grows endogenously...
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  - ... b/c firms invent **new** varieties of goods ([Romer, 1990](#))
  - ... b/c firms invent **better** varieties of goods ([Aghion and Howitt, 1992](#))
- but these were just models of “frontier” economies...
  - **today:** discuss how technology diffuses to the rest of the world

**Figure 7.1** Productivity and GDP per Capita

## How to close technology gaps?

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# How to close technology gaps?

- even w/ human capital, the best predictor of  $y_i/y_{US}$  is still  $A_i/A_{US}$
- why are some countries so far behind? how might they catch up?
- through the lens of the Romer (1990) model. . .
  1. a country can adopt ideas  $D_t$  from the frontier  $A_t$
  2. a country can import intermediate goods  $M_t$  from the frontier  $A_t$

(will always then have  $D_t + M_t \leq A_t$ )

## Final and intermediate good producers

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As in [Romer \(1990\)](#), varieties are symmetric and only require capital to produce:

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- frontier ideas  $A_t$  are treated as **exogenous** by adopting country
- parameter  $\zeta$  governs relative importance of  $A_t$  vs.  $D_t$  in ability to adopt
  - $\zeta \rightarrow 1$ : matters more to have more ideas to borrow/copy/purchase from frontier
  - $\zeta \rightarrow 0$ : matters more to have better domestic technology already (e.g., Internet)
  - *note*: could have been more general than Cobb-Douglas



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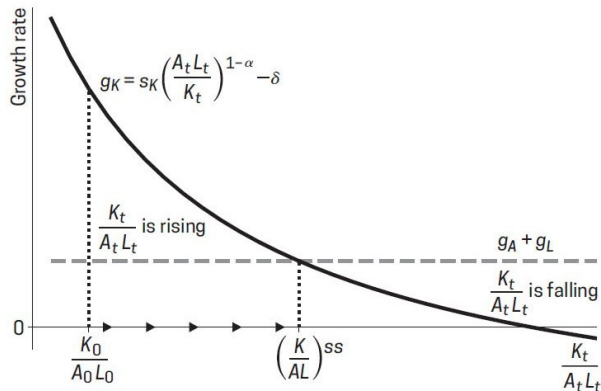
*how do we find it?*

Recall **similar expressions** from the Solow ( $g_K$ ) and Romer ( $g_A$ ) models...

$$g_K = s_I \left( \frac{A_t L_t}{K_t} \right)^{1-\alpha} - \delta \qquad g_A = \theta s_R^\lambda \left( \frac{L_t^\lambda}{A_t^{1-\phi}} \right)$$

## Use a phase diagram

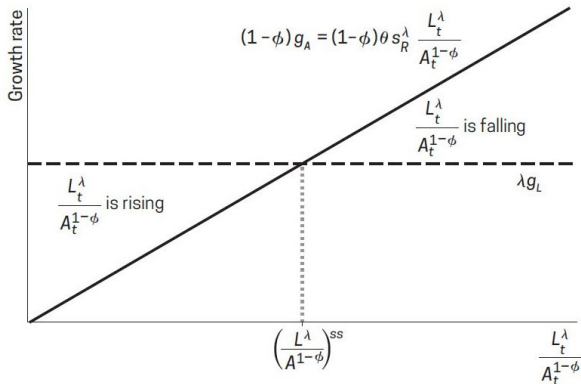
**Figure 2.2** The Dynamics of the  $K_t/A_tL_t$  Ratio



**NOTE:** The dark line plots the growth rate of capital,  $g_K$ , from equation (2.11) against the  $K_t/A_tL_t$  ratio. The growth rate of  $AL$ , which is  $g_A + g_L$ , is plotted in the dashed line. Everywhere the dashed line is above the solid line, the  $K_t/A_tL_t$  ratio must be falling, as  $g_K < g_A + g_L$ . Everywhere the dashed line is below the solid line, the  $K_t/A_tL_t$  ratio must be rising, as  $g_K > g_A + g_L$ .

# Use a phase diagram

**Figure 5.1** The Dynamics of Productivity Growth

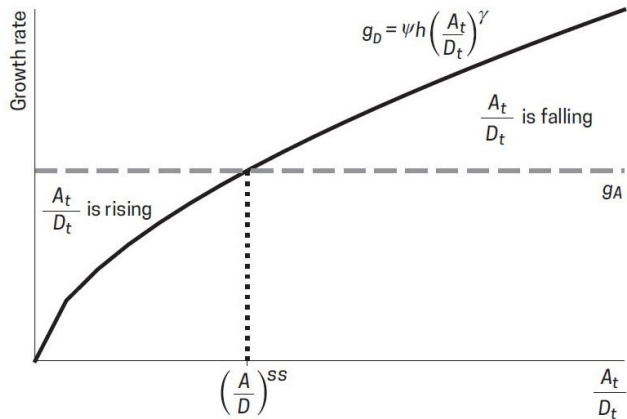


NOTE: The two curves plot the growth rate of the numerator and denominator of the ratio  $L_t^\lambda / A_t^{1-\phi}$ . The growth rate of the numerator is  $\lambda g_L$ , plotted as the horizontal line. The growth rate of the denominator is  $(1-\phi)g_A$ , which from equation (5.8) is increasing in the ratio. The economy reaches a steady state where the two lines intersect, and at that point the growth rate of productivity is  $g_A^{ss} = \frac{\lambda}{1-\phi} g_L$ .



## Use a phase diagram

**Figure 7.2** The Dynamics of Domestic Technology



NOTE: The growth rate of domestic technology,  $D_t$ , is from equation (7.10) and is rising with the ratio  $A_t/D_t$ . The growth rate of frontier technology,  $g_A$ , is taken as given in the domestic economy and does not change with  $A_t/D_t$ .

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$$\left(\frac{A}{D}\right)^{\text{ss}} = \left(\frac{g_A}{\psi h}\right)^{\frac{1}{\zeta}},$$

which means, along the BGP,

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The level of human capital enters twice:

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Recall our discussion of development accounting w/ human capital (Week 4 Slides 29–38)

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- **Hendricks and Schoellman (2018)**: check  $\Delta w_i$  for migrants  $\rightarrow h$  matters most

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$$\int_0^{D_t} z_{jt} dj = K_t \quad \rightarrow \quad \underbrace{D_t(z_t - x_t)}_{\text{exports}} = \underbrace{M_t x_t}_{\text{imports}}$$

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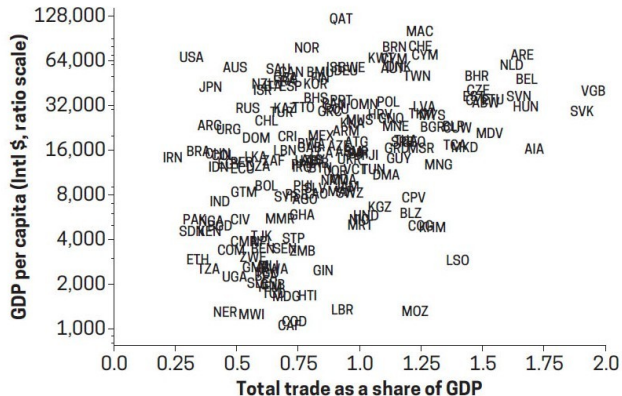
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- there is a positive level effect of **more imports**  $\rightarrow$  **why?**
  - diminishing returns to each variety ( $\alpha < 1$ )...
  - ... means it's better to have a small amount of more varieties  $\rightarrow$  **motive for trade**

# Trade openness is correlated with higher levels of GDP per capita

**Figure 7.3** Total Trade and GDP per Capita, 2019



SOURCE: Authors' calculations from Penn World Tables v10.0 (Feenstra, Inklaar, and Timmer, 2015).

NOTE: Total trade is the sum of exports and imports, divided by GDP, all in current national prices.  
GDP per capita is the purchasing power parity (PPP)-adjusted level.

$$\frac{\text{Imports}}{\text{GDP}} = \frac{M_t x_t}{Y_t} = \frac{M_t}{D_t + M_t} \frac{K_t}{Y_t}$$

- no countries in **lower right** → **closed** are poor / poor don't **trade**
- countries in **top left** tend to be at the tech. frontier → **invent** the varieties, don't **import** as many



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  - w/ production and physical capital accumulation as in the NGM
- but gaps—in levels *and* growth rates—can persist for a long time...
  - transitional dynamics from **physical capital** as in NGM
  - transitional dynamics from **technology** through adoption and trade

## References

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- Aghion, Philippe and Peter Howitt. 1992. "A model of growth through creative destruction." *Econometrica* 60 (2):323–351.
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